

**VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF ECONOMICS AND LAW
FACULTY OF INFORMATION SYSTEMS**



FINAL REPORT

TOPIC: MACHINE LEARNING TECHNIQUES FOR CUSTOMER CHURN: A COMPARATIVE STUDY

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Ho Chi Minh City, April 2023

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Ho Chi Minh City, April 2023

COMMITMENT

The research team hereby declares that the contents of topic “*Machine Learning Techniques for Customer Churn: A comparative Study*” were carried out by the team itself. In addition to the cited references, the information used in this study is self-researched and implemented by the group. The data is collected and processed honestly. If false information is discovered, our team takes full responsibility.

ABSTRACT

Customers are increasingly interested in the quality of service (QoS) that companies can offer them. Services offered by various vendors are not easily distinguished, increasing rivalry among companies to keep and improve QoS. Customer Relationship Management (CRM) tools are used to help businesses gain new consumers, maintain ongoing relationships with them, and improve customer engagement for increased revenue. CRM systems employ machine learning algorithms to evaluate personal and behavioral data from customers in order to provide organizations with a competitive edge by raising customer engagement rates. These algorithms can forecast which consumers are likely to churn and why. Predictions are used to create tailored marketing strategies and service offerings. This paper attempts to evaluate and assess the efficacy of various machine learning methods used to forecast churn. This proposed system can determine important factors affecting customer purchasing behavior in the finance - banking industry.

Keywords: customer churn, machine learning, stacking, voting.

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LIST OF ABBREVIATION

AUC	Area Under The Receiver Operating Characteristics.
CRM	Customer Relationship Management
KNN	K-nearest Neighbor
RF	Random Forest
ROC	Receiver Operating Characteristics
SVM	Support Vector Machine
XGBoost	eXtreme Gradient Boost

CHAPTER 1: OVERVIEW

1.1. Reasons for choosing the topic

Banks and financial institutions have a large number of clients in today's corporate climate. Banks offer services through a variety of methods, including ATMs, debit cards, credit cards, online banking, and so on. Consumers' awareness of service quality has grown dramatically as the number of consumers has grown dramatically. This fuels fierce rivalry among banks, resulting in a substantial rise in the dependability and quality of service provided by banks.

Customers are the foundation of any business's success and revenue, which is why businesses are becoming more conscious of the significance of obtaining customer satisfaction. Customer Relationship Management (CRM) aids marketing by identifying potential customers and establishing cost-effective partnerships with them. CRM is the process of analyzing client behavior in order to help businesses increase customer acquisition, retention, and revenue. CRM systems, as a result, use business intelligence and analytical models to find the most profitable set of customers and target them in order to achieve higher client retention rates. Based on an analysis of customers' personal, demographic, and behavioral data, these models can forecast customers who are likely to churn and provide customized and customer-oriented marketing efforts to increase customer happiness. The lifecycle of business – customer relationship includes four main stages: 1) identification; 2) attraction; 3) retention; and 4) development.

1.1.1. Customer identification

This aims to identify profitable customers and the ones that are highly probable to join organization. Segmentation and clustering techniques can explore customers' personal and historical data to create segments/sub-groups of similar customers.

1.1.2. Customer attraction

The identified customer segments/ sub-groups are analyzed to identify the common features that distinguish customers within a segment. Different marketing techniques can be used to target different customer segments such targeted advertising and/or direct marketing.

1.1.3. Customer retention

This is the main objective of CRM as retaining existing customers is at least 5 to 20 times more cost effective than acquiring new ones depending on business domains. Customer retention includes all actions taken by organization to guarantee customer loyalty and reduce customer churn. Customer churn refers to customers moving to a competitive organization or service provider. Churn can be for better quality of service, offers and/ or benefits. Churn rate is an important indicator that all organizations aim to minimize. For this sake, churn prediction is an integral part of proactive customer retention plan. Churn prediction includes using data mining and predictive analytical models in predicting the customers with high likelihood to churn/defect. These models analyze personal and behavioral customer data for tailored and customer-centric retention marketing campaigns.

1.1.4. Customer development

The main objective of this phase is to increase the amount of customer transactions for more profitability. For this sake, market basket analysis, customer lifetime value, up, and cross selling techniques are used. Market basket analysis tries to analyze customers' behavior patterns to maximize the intensity of transactions. Analyzing customer lifetime value can help identifying the total net income expected from customer. Up and/ or Cross selling include activities that increase the transactions of the associated services/ products.

Customer retention and churn prediction have been increasingly investigated in many business domains, including, but not limited to, telecommunication, banking, retail and cloud services subscriptions. Different statistical and machine learning techniques are used to address this problem. Many attempts have been made to compare and benchmark the used techniques for churn prediction. In a comparison between (Decision trees, Logistic regression and Neural Network) models was performed. The study found that neural network perform slightly higher than the other two techniques. Another comparison between a set of models against their boosted versions is discussed in. This study included two layer Back-Propagation neural network (BPN), Decision Trees, SVM and Logistic Regression. The study showed that both decision trees and BPN achieved accuracy 94%, SVM comes next with 93% while Logistic Regression

failed with accuracy 86%. Additionally, study showed 1-4% performance improvement in the boosted versions. In the study investigated the accuracy of different models (Multi-layer perceptron and Decision Tree (C5)). The study showed that Multi-layer perceptron achieves accuracy of 95.51%, which outperforms C5 decision tree 89.63%. Most of comparisons in the literature did not consider a study that covers the various categories of learning techniques.

The bulk of the models applied for churn prediction fall into one of the following categories:

- Regression analysis,
- Decision tree-based,
- Support Vector Machine,
- Bayesian algorithm,
- Instance – based learning,
- Ensemble learning,
- Artificial neural network,
- Linear Discriminant Analysis

The main goal of this study compares the efficiency and effectiveness of the model in the literature and predicting customer churn rate through different commonly used algorithms. The selected models are:

- Logistic Regression,
- Decision Tree,
- Support Vector Machine,
- Instance – based learning: K-nearest Neighbors (KNN),
- Ensemble learning: Random Forest (RF), Extreme Gradient Boosting (XGBoost), Stacking, Voting.

The research topic is “*Machine Learning Techniques for Customer Churn: A comparative Study*”. At the same time, this research also provides the basis for future studies in the same field.

1.2. Objectives of the study

1.2.1. General Objective:

Comparing and researching about Customer Churn, especially the bank. We use python as the main programming language and some technology software such Python, Excel,... to serve the research's goals. Conducting data analysis from ABC Multistate bank data to create stories about the churn and related the characteristics of customer, helping the bank and its stakeholders spot the bright spots, understand the bank's situation clearly, thereby assisting the bank in setting up a better strategies in the future of customer services.

1.2.2. Specific Objective.

- Analyzing and benchmarking the performance of the models in literature
- Giving levels of exactly predicting customer churn through the used algorithms.
- Giving governance implications to help bank reduce customer churn rate, improving customer services and attracting new customers

1.3. Subjects and scope of the study

1.3.1. Subjects of the study

ABC Multistate bank (intercorporated as Arab Banking Corporation B.S.C) is an international bank headquartered in Manama, Kingdom of Bahrain. Its network spreads across five continents, covering countries in the Middle East, North Africa, Europe, the Americas and Asia.

In 2021, more than doubling its customer base, growing deposits by 75% and achieving average daily transaction volumes seven times that of 2020. By focusing on our digital mobile-only bank, ila, in Bahrain which introduced multiple new features during the year-building out multi-currency accounts and launching the much-awaited ila credit cards and digital payments features, such as Apple Pay, Fitbit Pay and Garmin Pay.

Reflecting this success, ila was named “Best Consumer Digital Bank” at the Global Finance Innovators Awards in 2021 and received the “Best Digital Banking App” award.

In this study, we will use only the customer data of account holders at ABC Multinational Bank, containing information related to the account holders data at ABC Multinational Bank such as information of customers, account balance of customers as well as products, etc.

1.3.2. Scope of the study

1.3.2.1. Theory

- Learning about customer loyalty, data analysis methods and algorithms, finding anomalies between customer accounts.
- Learning about methods for predicting future model accuracy.

1.3.2.2. Empire

- Using statistical knowledge and methods to find anomalies from the dataset.
- Running algorithms to select the best method of accurate prediction.
- Developing research based on combining multiple models to deliver optimal results.

1.4. Methodology:

The research paper is based on the following methods.

1.4.1. Theoretical research:

- Learning definition, theories, causes, methods for predicting customer churn accuracy from available dataset and related research.
- Learning predictive methods and suggesting governance implications for current issues.

1.4.2. Empirical research:

Data collection, discovery analysis and data preprocessing, proposing a level analysis model that accurately predicts customer churn at the bank.

1.5. Structure of the study

CHAPTER 1: Overview

The research team introduces overview about study such as: reasons for choosing the topic, objectives of the study, scope and subjects of the study, methodology, structure and value,desired outcome of the study.

CHAPTER 2: Theoretical Basis

Report will present the theoretical basis and methods are used in team's study. It also provides an overview of the history of previous studies.

CHAPTER 3: Customer churn prediction model

In this chapter, descriptions of data, preprocessing, attribute selection, attribute building, new customers and current customers, how to combine machine learning methods in model building will be detailed.

CHAPTER 4 : Empirical and Proposed results.

From the model built in chapter 3, the team will evaluate the performance and results of the model with the indexes used for the classification problem as well as the intergration ability model to form a prediction of level of accuracy system for business.

CHAPTER 5 : Conclusion and Future Works

From results received, the team will give relevant conclusions and provide directions for future research expansion.

1.6. Value and desired outcome of the study

1.6.1. Value of the study

This study was conducted with aim of developing a model that predicts customer churn in the business (specifically bank) to offer appropriate solutions for the next action to retain customers. In addition, the accuracy of the models will help contribute to increasing operational efficiency in the business of customer services, ensuring to bring satisfaction to customers, especially loyal customers of the business.

1.6.2. The desired outcome of the study

The research results will support businesses to come up with appropriate strategies and solutions to reach new customers as well as retain loyal customers. Indirectly contributing to serving the developed country's economy, especially helping businesses that do not have good predictive models can refer to complete them. Since then, both businesses and customers will benefit together.

CHAPTER 2: THEORETICAL BASIS

Chapter 2 presents the theories and methods the study of team surveyed and used during the implementation of the topic. At the same time, related studies before that is also mentioned an overview of the research situation.

2.1. Definition of customer loyalty

Customer loyalty has been the popular topic among the business workplaces (Leong et al 2012) and recent years have shown a growing interest in customer loyalty (Kuusik 2007) so has been widely researched in the domain of marketing (BOSE and RAO 2011). Loyalty is a positive belief in the value that a company provides, leading to increased purchases over time. “Loyalty is a positive belief, generated over the course of multiple interactions, in the value that a company and its products or services provide, which leads to continued interactions and purchases over time” (Oracle Corporation, 2005).

Szűts & Tóth (2008) expressed that ‘customer loyalty’ is a customer's demonstration of faithful adherence to an institution or merchant despite the occasional error or indifferent service. In this view, customer loyalty is an attitude or behavior that customers explicitly vocalize or exhibit.

Generally speaking, customer loyalty means the intention of repurchasing products and services, and this is the goal of industry or loyalty is a deeply held commitment to re-buy or re-patronize a preferred product or service consistently in the future, thereby causing repetitive same-brand or same -brand set purchasing. So customer loyalty is seen as one of the key factors of a company’s success.

2.2. Machine Learning

Machine learning is a broad subject with applications in information technology, statistics, probability, artificial intelligence, psychology, neuroscience, and many other fields. Machine learning can address issues by simply creating a model that is a decent representation of a given dataset. Machine learning has progressed from training computers to imitate the human brain to a broad science that generates basic statistical computational theories of learning processes. The goal of machine learning is to create

methods that enable computers to learn. Learning is the process of discovering statistical regularities or other data trends. Machine learning algorithms are designed to mimic the human method to learning a job.

Supervised learning: the various algorithms generate a function that maps inputs to desired outputs. One standard formulation of the supervised learning task is the classification problem: the learner is required to learn (to approximate the behavior of) a function which maps a vector into one of several classes by looking at several input - output examples of the function.

Unsupervised learning: models a set of inputs: labeled examples are not available.

2.3. Supervised learning

In supervised learning algorithms, the classes are predetermined. These classes are created in a manner of finite set, defined by the human, which in practice means that a certain segment of data will be labeled with these classifications. The task of the machine learning algorithm is to find patterns and construct mathematical models. These models are then evaluated based on the predictive capacity in relation to measures of variance in the data itself. It is also useful to make difference between two main supervised models: *classification models* (classifiers) and *regression models*. Regression models map the input space into a real-value domain. The classifiers map the input space into predefined classes. There are many alternatives for representing classifiers, for instance, support vector machines, decision trees, probabilistic summaries, algebraic function, etc. Along with regression and probability estimation, classification is one of the most studied models, possibly one with the greatest practical relevance. The potential benefits of progress in classification are immense since the technique has great impact on other areas, both within Data Mining and in its applications.

2.3.1. Background

The learning process in a simple machine learning model is divided into two steps: training and testing. In training process, samples in training data are taken as input in which features are learned by learning algorithm or learner and build the learning model. In the testing process, learning model uses the execution engine to make the prediction for the test or production data. Tagged data is the output of learning model which gives the final prediction or classified data.

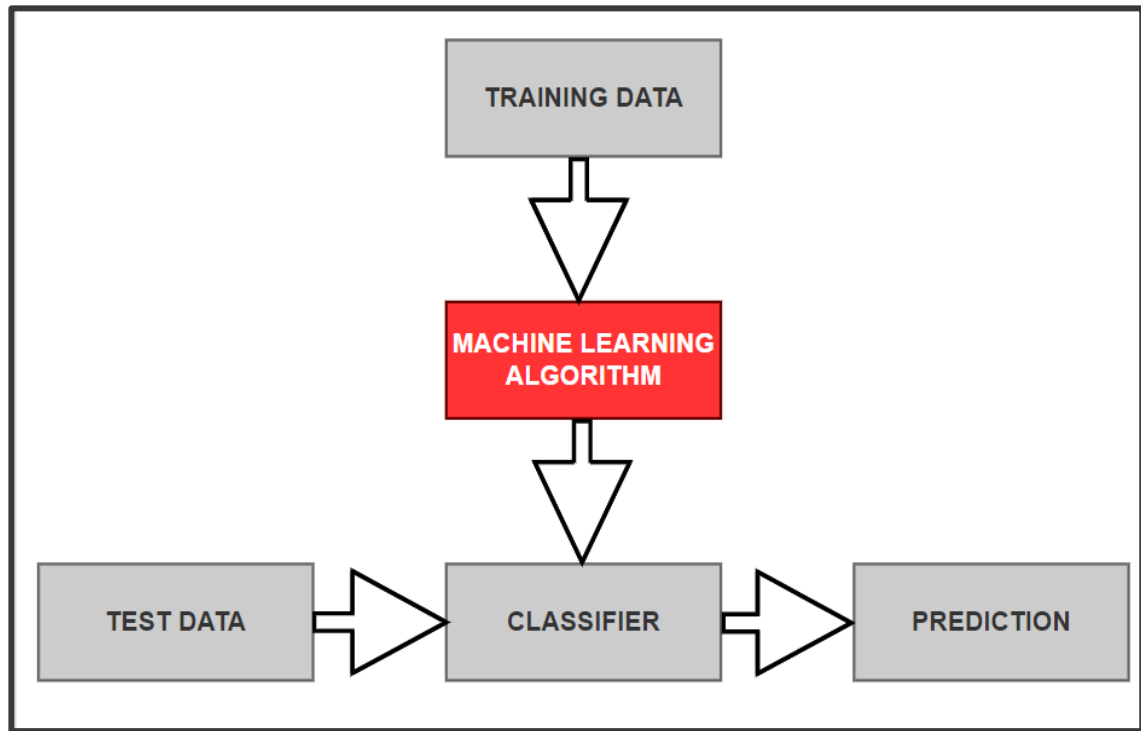


Figure 1: Supervised process

(Resource: Faculty of Information and Communication Technologies, Partizanska bb, 7000 Bitola, Macedonia.)

Supervised learning (Figure 1) is the most common technique in the classification problems, since the goal is often to get the machine to learn a classification system that we've created.

The supervised learning tasks are divided into two categories: classification and regression. In classification, the label is discrete, while in regression, the label is continuous.

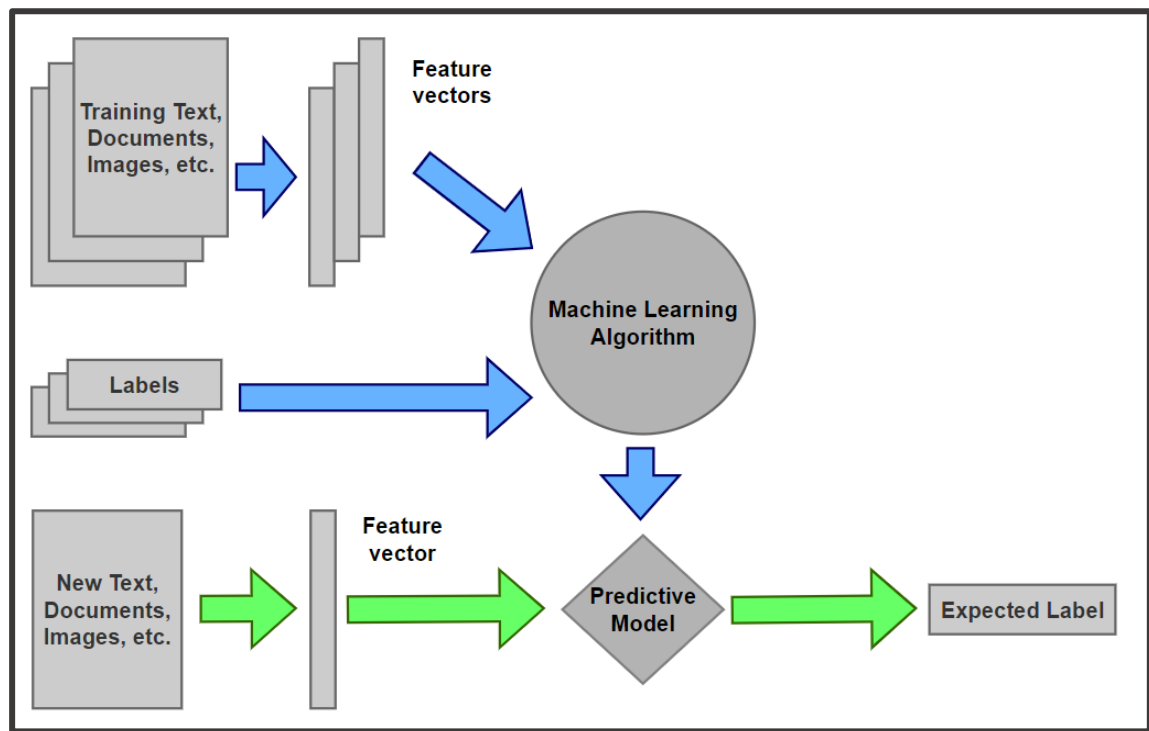


Figure 2: Supervised learning model

(Resource: Faculty of Information and Communication Technologies, Partizanska bb, 7000 Bitola, Macedonia.)

As shown on Figure 2, the algorithm makes the distinction between the observed data X that is the training data, in most cases structured data given to the model during the training process. In this process, the supervised learning algorithm builds the predictive model. After its training, the fitted model would try to predict the most likely labels for a new set of samples X in the testing set. Depending on the nature of the target y , supervised learning can be classified:

If y has values in a fixed set of categorical outcomes (integers), the task to predict y is called classification.

If y has floating point values, the task to predict y is called regression.

2.3.2. Algorithms

2.3.2.1. Logistic Regression.

Logistic regression works by extracting some set of weighted features from the input, taking logs and combining them linearly, which means that each feature is multiplied by a weight and then added up.

The most important difference between naive Bayes and logistic regression is that the logistic regression is a discriminative classifier while the naive Bayes is a generative classifier.

Logistic regression is a type of regression that predicts the probability of occurrence of an event by fitting data to a logistic function. Just as many form of regression analysis, logistic regression makes use of several predictor variables that may be numerical or categorical.

The logistic regression hypothesis is defined as:

$$h_{\theta}(x) = g(\theta^T x) \quad (1)$$

Where the function g is sigmoid function defined as:

$$g(z) = \frac{1}{1+e^{-z}} \quad (2)$$

The sigmoid function has special properties that result the values in range $[0,1]$, as visualized on Figure 3.

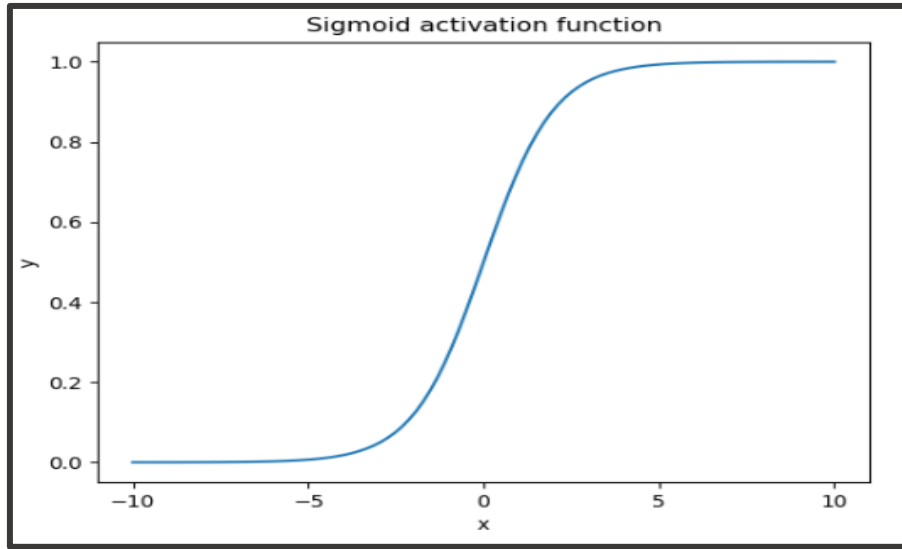


Figure 3: Visual representation of the Logistic Function

(Resource:Internet.)

The cost function for logistic regression is given as:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \left[-y^i \log(h_{\theta}(x^i)) - (1 - y^i) \log(1 - h_{\theta}(x^i)) \right] \quad (3)$$

To find the minimum of this cost function, in machine learning we will use a builtin function called `fmin_bfgs`², which finds the best parameters θ for the logistic regression cost function given a fixed dataset (of x and y values). The parameters are the initial values of the parameters that need to be optimized and a function that when given the training set and a particular θ , computes the logistic regression cost and gradient with respect to θ for the dataset with x and y values. The final θ value will be used to plot the decision boundary of the training data.

2.3.2.2. Decision Tree.

A decision tree is a recursive division of the instance space that symbolizes a classifier. The decision tree is made up of nodes that make a so-called root tree, which means that it is a distributed tree with no inbound edges and a fundamental node named root. Every other node has precisely one inbound edge. The node with outward edges is known as an interior node or a test node. The remaining components are referred to as branches. Each test node in a decision tree divides the instance space into two or more sub-spaces based on a finite function of the input values. In the most basic situation, each test examines a single attribute, and the instance space is divided according to the value of the attribute. In case of numeric attributes, the condition refers to a range.

Each leaf is assigned to one class, which reflects the best goal value. The leaf may contain a chance vector indicating the likelihood of the target trait having a specific value. The cases are categorized by navigating them from the tree's trunk down the leaf, based on the results of the tests along the way. Figure 3 depicts a basic application of the decision tree. Each node is designated with the attribute being tested, and its branches are labeled with the numbers associated with that attribute.

Given this classifier, the analyst can predict the response of come potential customer and understanding the behavioral characteristics of the entire potential customers's pollution.

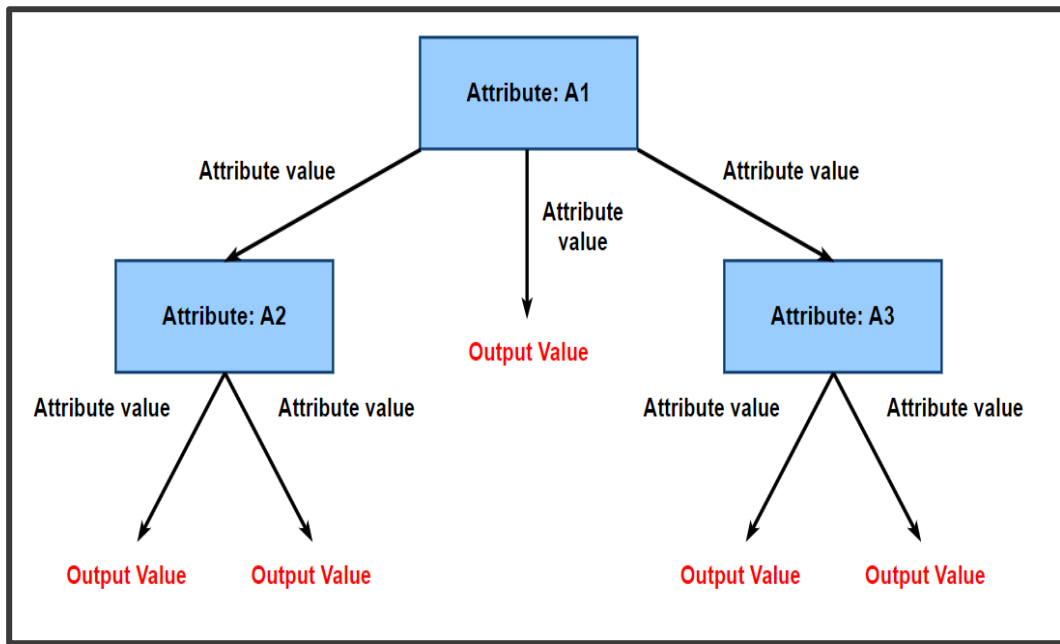


Figure 4: Decision tree example

(Resource: Internet)

In case of numeric attributes, decision trees can be geometrically interpreted as a collections of hyper planes, each orthogonal to one of the axis. Decision-makers prefer less complex decision trees, since they may be considered more comprehensive.

2.3.2.3. Support Vector Machine

This algorithms refers to the group of boundary techniques: it determines classes with use of area boundaries. The technique is based on the concept of solution planes. A solution plane divides objects of different classes. In high-dimension spaces instead of straight lines there should be considered hyperplanes of one dimension less than the one of the assumed space. In R3, for example, a hyperplane is a two dimensional plane.

The support vector technique searches for samples, located on class boundaries (at least two), i.e. support vectors, and decides the problem of finding the division a set of objects to classes with use of a linear decision function. The support vector technique builds classification function $f(x)$ as follows:

$$f(x) = \text{sign}((w, s) + b) \quad (4)$$

Where $\langle , \rangle$ – scalar product,

w – normal (perpendicular) vector to a dividing hyperplane,

b – auxiliary parameter equal to the absolute module of a distance from a hyperplane to the origin of coordinates. If a parameter b equals zero, a hyperplane passes through the origin of coordinates.

Objects with $f(x)=1$ get into the first class, but objects with $f(x) = -1$ - into another one.

From the point of classification accuracy the best straight line to be selected is the one maximally distant from each class. Such straight line (in general - a hyperplane) is identified as an optimal dividing hyperplane. The problem consists in selection of w and b , maximizing this distance.

In case of non-linear division the approach of support vector machine adaptation is available. It is necessary to input the space of R_n indicators into the space H of a larger dimensionality using the function of the form: $\varphi = R_n \rightarrow H$.

Hence decision of the problem reduces to a linear divisible case, i.e. a dividing classification function is searched again as:

$$f(x) = \text{sign}((w, \phi(x)) + b) \quad (5)$$

Also possible another option of data conversion – conversion to polar coordinates:

$$\begin{cases} x_1 = r \cos(\phi) \\ x_2 = r \sin(\phi) \end{cases}$$

2.3.2.4. Instance – based learning: K-nearest Neighbor

The reason for prediction using the K-nearest Neighbor method is based on the assumption that objects of neighbors have similar prediction values. The basic idea of the nearest neighbor algorithm is to find k points nearest to the unknown sample in the multidimensional space R_n , and judge the class of the unknown sample according to the categories of the k points. These k points are the k -nearest neighbors of the unknown samples. The algorithm assumes that all instances correspond to points in n dimensional

space. The nearest neighbor of an instance is defined according to the standard Euclidean distance. Let the eigenvector of x be:

$$\langle a_1(x), a_2(x), \dots, a_n(x) \rangle$$

Where $ar(x)$ represents the r th attribute value of instance x . The distance between the two instances x_i and x_j is defined as $d(x_i, x_j)$, where:

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^n [ar(x_i) - ar(x_j)]^2} \quad (6)$$

In nearest neighbor learning, the discrete object classification function is $f: R^n \rightarrow V$ where V is a finite set $\{v_1, v_2, \dots, v_s\}$, i.e. different sets of categories. The selection of the nearest neighbor k value is based on the number and degree of dispersion in each type of sample, and different k values can be selected for different applications.

If the number of sample points around the unknown sample is small, the area covered by the k points will be large, and vice versa. Therefore, the nearest neighbor algorithm is susceptible to noise data, especially the effects of isolated points in the sample space. The root cause lies in the basic KNN algorithm, in which the positions of the k nearest neighbor samples of the sample to be predicted are equal. In a natural society, usually an object is affected by its neighbors, and the closer the object is, the more influence it has.

2.3.2.5. Random Forest

Random Forest is a statistical algorithm modeled by machine and a method by Leo Breiman (2001). It can be used for both Classification and Regression. It is also flexible and easy to use algorithm. A forest consists of many trees. It is said that the more trees it has, the more robust a forest is. RF creates decision trees on randomly selected data samples, gets prediction from each tree and selects the best solution by means of voting (L.Breiman, 2001). It also provides a pretty good indicator of the feature importance.

→ Prediction is based on combination of many decision trees, by taking the average of all individual predictions.

Each tree in RF is simple but random.

Each tree is grown differently, depending on the choices of the attributes and training data.

Currently, RF is one of the most popular and accurate methods in application in various fields by a large number of properties, capable of estimating the importance of attributes, often with high accuracy in Classification (or Regression).

RF can be implemented easily and efficiently. Besides that, it also can work with problems of very high dimensions, rarely in cases of overfitting. However, little is known about its theoretical properties, we cannot rigorously prove that it can best avoid overfitting or recirculation. So far, Mathematicians have only come up with a few analyses to support a few ideas about random forest to show that model works.

2.3.2.6. *Extreme Gradient Boosting*

XGBoost is a boosting algorithm belonging to supervised learning, which is an ensemble algorithm based on gradient boosted trees. It integrates predictions of “weak” classifiers (tree model) to achieve a “strong” classifier (tree model) via a serial training process. It can avoid over-fitting by adding a regularization term. Parallel and distributed computing makes the learning process faster to give a quicker modeling process. Figure 5 shows a schematic diagram of the computational process of XGBoost and y_i appeared in the process is calculated by Equation 7.

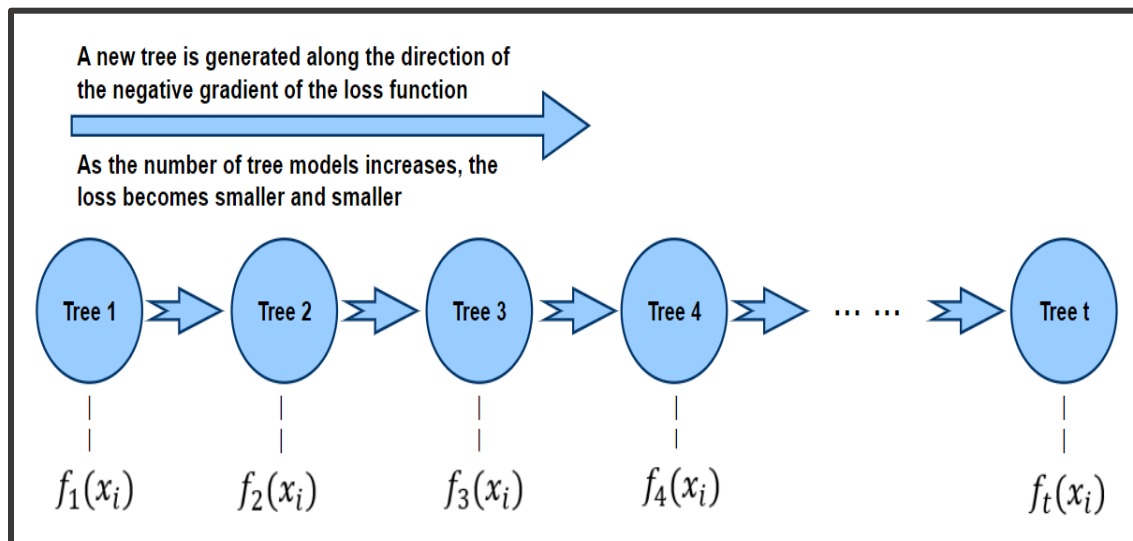


Figure 5: A schematic diagram of the computational process of XGBoost

(Resource: Earth and Environmental Science 113.)

$$\hat{y}_i^{(t)} = \phi(x_i) = \sum_{k=1}^k f_k(x_i) \quad (7)$$

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i) \quad (8)$$

Where, $\hat{y}_i^{(t)}$ is the final tree model; $\hat{y}_i^{(t-1)}$ is the previously generated tree model; $f_t(x_i)$ is the newly generated tree model, and t is the total number of base tree models

For the XGBoost algorithm, both depth and number of trees are important parameters. The problem of finding the optimal algorithm was changed into finding a new classifier that can reduce the loss function, with the target loss function shown in Equation 3,

$$Obj^{(t)} = \sum_{i=1}^t L(y_i, \hat{y}_i^{(t)}) + \sum_{i=1}^t \Omega(f_i) \quad (9)$$

Where, y_i is the actual value; $\hat{y}_i^{(t)}$ is the predicted value; $L(y_i, \hat{y}_i^{(t)})$ is the loss function and $\Omega(f_i)$ is the regularization term.

Substituting Equation 2 into Equation 3 and then following some deduction steps (can be found in), Equation 4 could be obtained,

$$Obj^{(t)} = \sum_{i=1}^t L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) + constant \quad (10)$$

The final target loss function was then converted into Equation 5, and the model was then trained according to this target loss function.

$$Obj^{(t)} = \sum_{i=1}^t [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (11)$$

Where, $g_i = \partial_{y_i^{(t-1)}} l(y_i, y_i^{(t-1)})$ and $h_i = \partial_{y_i^{(t-1)}}^2 l(y_i, y_i^{(t-1)})$ are the first and second order gradient statistics on the loss function.

The regularization term $\Omega(f_i)$ is calculated by Equation 6 to reduce the model's complexity and also improve its usability to another dataset.

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2 \quad (12)$$

where, T is the number of leaves; ω is the weight of the leaves; λ and γ are coefficients, with default values set as $\lambda=1, \gamma=0$.

The XGBoost algorithm can accept both continuous variables and discrete variables as inputs but the output variable has to be discrete, including binary variables. In this study, the XGBoost algorithm was run in Python 3.6. When using the XGBoost algorithm, Z-statistic is often used for testing the significance of each independent variable, with p-value given at 95% confidence interval. It is calculated and given by the computational package after running the XGBoost algorithm.

2.3.3. Stacking

Stacking (also known as Stacked Generalization) is a distinct concept. Stacking is used to investigate a universe of various models for the same issue. The notion is that you can approach a learning problem with various kinds of models that can learn a portion of the problem but not the entire problem space. So you can create numerous distinct learners and use them to create an intermediate prediction, one for each taught model. Then you add a new model that learns the same goal from the intermediate forecasts.

In the standard stacking algorithm shown in Figure 6, n different subsets of the training data set are created by using stratified sampling with replacement in which the relative proportion of the different classes is maintained in all the subsets. Each subset of the training set used to determine the performance of the classifiers on the training set. A meta classifier in the form of relative weight for each classifier is created by assigning a weight to a classifier that is proportional to its performance.

When evaluating an instance from the test set, every classification algorithm in WEKA gives a class distribution vector for that instance that gives the probability of that particular instance belonging to a given class. We can represent the class distribution vector over c classes for the j th classifier by a $1 \times c$ vector as follows:

$$\Delta_j = [\delta_{1j}, \delta_{2j}, \dots, \delta_{cj}] \quad 1 \leq j \leq n \quad (13)$$

where,

$$0 \leq \delta_{ij} \leq 1 \quad \forall 1 \leq i \leq c$$

$$\sum_i \delta_{ij} = 1$$

The class distribution vectors for the n classifiers can then be represented by an $n \times c$ matrix as follows:

$$\mathbf{\Delta} = [\Delta_1, \Delta_2, \dots, \Delta_n]^T \quad (14)$$

The meta-classifier creates a weight distribution vector that gives relative weight to different classifiers. The weight distribution vector over n classifiers is represented as follows:

$$\theta = [\theta_1, \theta_2 \dots \theta_n] \quad (15)$$

where,

$$0 \leq \theta_j \leq 1$$

$$\sum_j \theta_j = 1$$

Given the class distribution matrix and the weight distribution vector, the meta-classifier evaluates each instance of the test set by using the following $1 \times c$ class distribution vector:

$$j = \theta \cdot \mathbf{\Delta} = [\delta'_1, \delta'_2 \dots \delta'_c] \quad (16)$$

where,

$$\delta'_i = \sum_j \theta_j \delta_{ij}$$

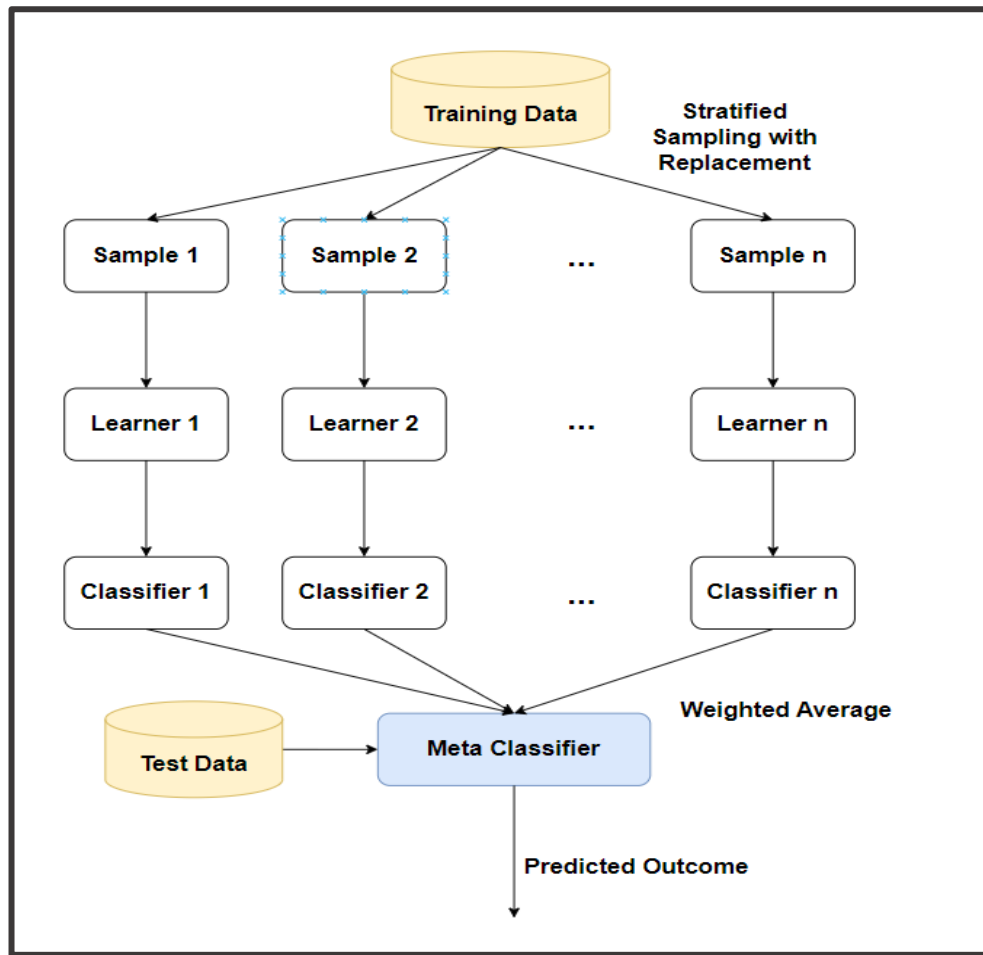


Figure 6: Standard Stacking ensemble learning

(Resource: Journal of International Technology and Information Management 2014 23(1), 1-12.)

As mentioned above, in the standard stacking algorithm the meta-classifier weight distribution vector Θ is created by assigning a weight to a classifier that is proportional to its performance. In the next section we discuss using a genetic algorithm to learn the weight distribution vector

2.3.4. Voting

Voting is important for ultrareliable systems that are based on the multi-channel computation paradigm. Voting is required whether the multiple computation channels consist of redundant hardware units, diverse, identical hardware and software with diverse data, or any other combination of hardware/program/data redundancy and/or diversity. Depending on the data volume and the frequency of voting, hardware or software voting schemes can be appropriate. Low-level voting with high frequency requires hardware voters whereas high-level voting on the results of fairly complex

computations can be performed in software without serious performance degradation or overhead.

The use of voting for obtaining highly reliable data from multiple unreliable versions was first suggested in the mid 1950s. Since then the concept has been practically used in fault-tolerant computer systems and has been extended & refined in many ways, eg.

- Reliability modeling of voting schemes by considering compensating errors.
- Handling of imprecise or approximate data
- Combination with standby or active redundancy
- Reconfigurable voting with declining replication factor upon detected failures
- Voting on digital “signatures” obtained from computation states to reduce the amount of information to be voted on
- Dynamic modification of vote weights based on a priori reliability data.

2.4. Evaluation measures.

In order to evaluate classifiers performance in churn prediction for different schemes with their appropriate parameters. We use the measures of “Precision”, “Recall” and “ Accuracy”, calculated from the contents of the confusion matrix, shown in Table X. True positive and false positive cases are denoted as TP and FP, respectively, while true negative and false negative cases are denoted as TN and FN.

CONFUSION MATRIX	Predicted class		
		Churners	Non-churners
	Churners	TP	FN
	Non-churners	FP	TN

Table 1: Confusion matrix
(Resource: Do-it-yourself team)

Where,

TP: is the total number of customers correctly identified as churn.

FP: is the total number of customers incorrectly identified as churn.

TN: is the total number of customers correctly identified as no-churn.

FN: is the total number of customers incorrectly identified as no-churn.

Precision is the proportion of the predicted positive cases that were correct and is calculated from the equation:

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

Recall is the proportion of positive cases that were correctly identified and is calculated from the equation:

$$Recall = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

Accuracy is the proportion of the total number of predictions that were correct and is calculated from the equation:

$$Accuracy = \frac{True\ Positive + True\ Negative}{True\ Positive + False\ Positive + True\ Negative + False\ Negative}$$

For ROC curve, a layered model is considered effective when False Positive Rate is low and True Positive Rate is high, i.e. there is a value on the ROC curve that is closer to the point with coordinates (0.1), the more the curve needs to be with, the better the model efficiency. AUC is considered an accompanying parameter in the evaluation of layered models. This is an indicator that measures the layering ability of the based model ROC Curve, which is the portion of area that lies under the ROC curve and is surrounded by a complete axis, between [0.1], the greater this value indicates the model the better.

Similar to the ROC curve, changing a threshold and observing the value of Precision and Recall create a Precision-Recall curve. With the Precision - Recall Curve, AUC is also known as the Average Precision (AP). Suppose there are N thresholds for calculating Precision and Recall, at each observation threshold for a pair of Precision results, Recall is P_n, R_n , with $1 \leq n \leq N$. The Precision-Recall curve is a straight line

that connects points with coordinates (R_n, P_n) together. AP is determined by the formula (17):

$$Ap = \sum_n (R_n - R_{n-1}) P_n \quad (17)$$

2.5. Identification of customer churn by machine learning method

In Sahar F.sabbbeh's (2018) comparative study "Machine Learning Techniques for Customer Retention," the author discovered that customers have become more interested in the quality of service provided by different vendors, which increases competition between organizations to maintain and improve their quality of service. As a result, using machine learning algorithms to forecast consumers who are likely to churn and the causes for churn is critical. The author used ten analysis methods and produced accurate results: 1) Random forest and ADA boost: 96%; 2) Multi-layer perceptron and SVM: 94%; 3) Decision tree: 90%; 4) Nave Bayesian: 88%; and 5) Logistic regression and linear discriminant analysis (LDA): 86.7%. We discovered that models with forecast results less than 90% are not yet effective, and models with great precision are not yet effective in detecting client turnover.

In addition, Manas Rahman's (2020) research "Machine Learning based customer churn prediction in banking" has the same aim of predicting customer loss. The author used the K-nearest neighbor (KNN), SVM, Decision tree, and Random forest algorithms to forecast, and the precision after oversampling is as follows: 1) KNN: 81.37%, 2) SVM: 70.36%, 3) Decision tree: 91.98%, and 4) Random forest: 95,74%. As a consequence, the study team discovered that the forecast results of KNN and SVM models are quite poor, with an increased risk of mistakes.

CHAPTER 3: CUSTOMER CHURN PREDICTION MODEL

In this chapter, the research team presents the process of analyzing and building models from the theoretical basis and data collected by the team during previous implementation. The collected data is then fed into data exploration analysis to comprehend the empirical dataset. Besides, data preprocessing will be done to clean and shrink the dataset size. From there, as a basis for making a customer churn prediction model on a combination of the above-mentioned algorithms.

3.1. Analysis and modeling process

3.1.1. Target identification and analysis

The request for analysis is a direction to help the research have a specific and more meaningful direction. Understanding the problem to be analyzed is the most important aspect, laying the foundation for proper research. Understanding the analysis objectives helps identify important indicators to monitor, evaluate, and draw conclusions directly related to the core issue, as well as providing a reasonable explanation for behavior and trends from the data. This process requires certain knowledge of the data research field and the connection between parties to understand the essence of the link between variables in the data set.

3.1.2. Proposed model

After researching and discussing, the team decides to propose the following model:

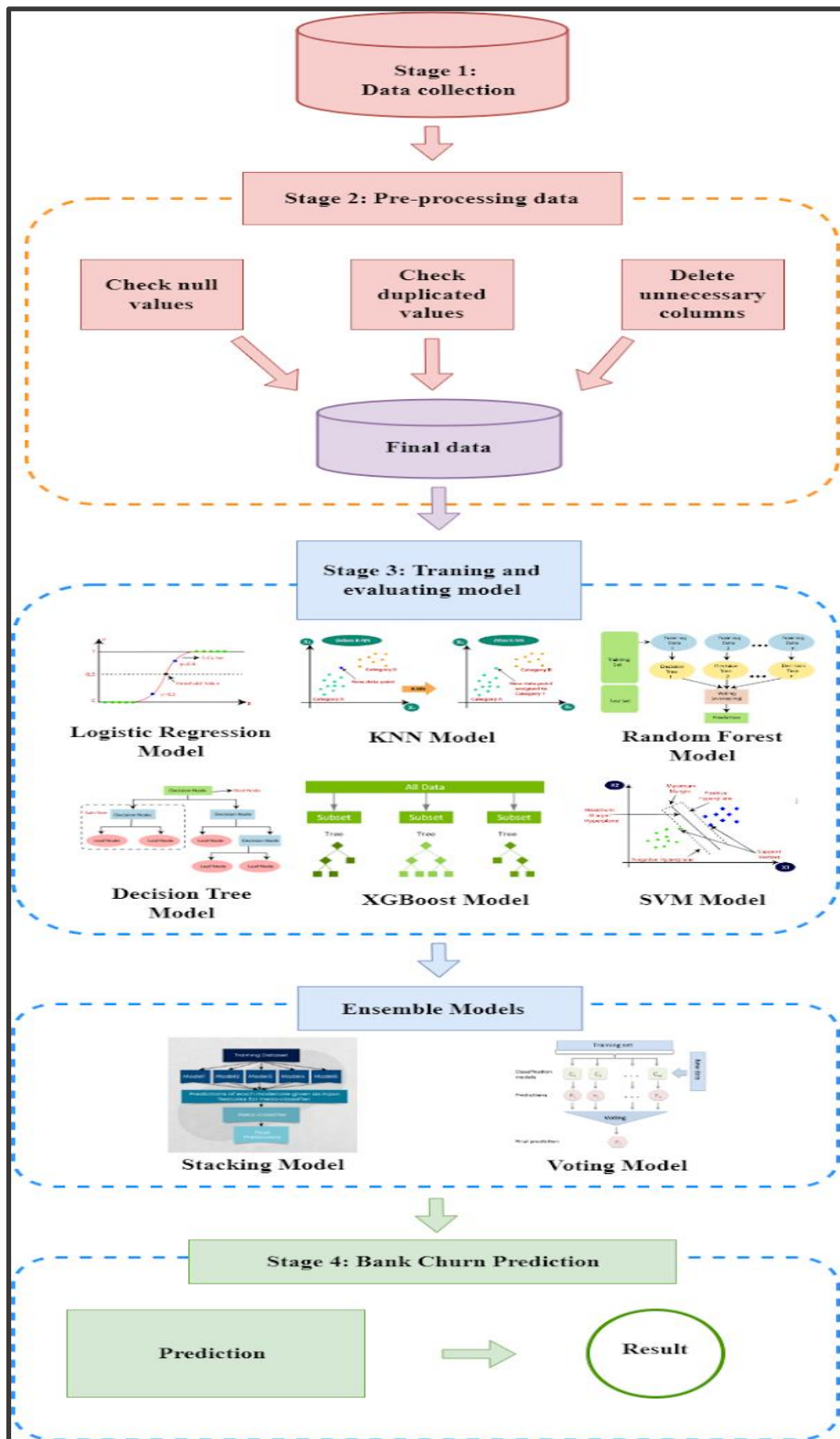


Figure 7: Proposed model
(Resource: Do-it-yourself team)

3.1.3. Data collection

Data is the main raw material for solving research questions. From the identified problem, the attributes of the data will be selected based on their relevance to the issue as much as possible. The data source is divided into two types: direct sources collected by the company from customers who purchase products or use the company's services, or indirect sources purchased from another party other than the company.

Data can be collected in structured form (tabular data) or unstructured form (non-tabular data such as images, audio). For this research topic, the data used is structured data with variables and categorical variables. Each type of variable will be processed differently before becoming input data for the model.

3.1.4. Exploratory data analysis

This stage provides an overview of the analyzed data set through statistical analysis and data visualization to understand the size of the data, the current number of attributes, the amount of duplicated or missing data, identifying outliers, and finding appropriate methods to handle them. It also includes calculating statistical indices, distribution characteristics of each attribute, and the correlation between attributes.

3.1.5. Data pre-processing

3.1.5.1. Cleaning data

The occurrence of errors such as duplication, incorrect or missing information cannot be avoided in the provided data set. To ensure accuracy for the conclusions drawn from analysis as well as building models with high accuracy, the data cleaning process is extremely important.

Data cleaning involves operations such as removing duplicate observations, handling missing data, correcting data to the correct format, especially time data types, checking for syntax errors, spelling errors, and examining whether outliers are due to input errors or other reasons.

3.1.5.2. Attribute transformation

This is the process of transforming the original values into data that is suitable for the research problem. The transformed attributes often have a higher correlation with the dependent variable, thereby improving the accuracy of the model. Commonly used methods for attribute transformation include normalization and changing the data distribution through scaling techniques such as Mean normalization, Standardization, Unit length scaling, and Minmax scaling.

3.1.5.3. Creating new attributes

In the process of data analysis, it is not mandatory to create new attributes, and they often arise in the optimization process to improve the accuracy of the model. There is no general standard for creating new attributes, which mainly depends on experience and creative thinking based on the characteristics of the analyzed data set.

3.1.5.4. Feature selection

Feature selection is often used in datasets with many variables that have a significant impact on the model's classification ability. Selecting the right features is crucial to avoid overfitting, where the model performs well on the training set but poorly on the testing set. Additionally, a streamlined dataset provides simplicity and efficiency in the modeling process, reducing complexity and facilitating faster and easier computations.

3.1.6. Building analytical model

The process of building a model typically involves four stages: model training, cross-validation and parameter tuning to improve the model:

3.1.6.1. Training and Testing

The initial dataset will be divided with a higher proportion for the training set and the remaining part for evaluating the model's predictive ability. In this case, the test set plays a role as a sample of the population, which is the real data of the problem being solved. Therefore, the results of the model on the test set are a reliable measure of the model's effectiveness.

3.1.6.2. Cross-validation

Cross-validation helps to evaluate and compare the predictive ability of machine learning algorithms based on the data set divided into multiple validation sets, while preventing the overfitting phenomenon.

3.1.6.3. *Model tuning*

The tuning process involves modifying the parameter set of the algorithm to improve the prediction results. The model is built sequentially based on these parameter sets, and the results are compared to select the parameter set with the highest performance.

3.1.7. Evaluate model and conclusions

Depending on the initial analysis goal, there will be corresponding evaluation metrics. For the problem of classifying authors' groups being studied, commonly used metrics include accuracy, precision, recall, ROC and AUC score.

3.2. Model overview

The process begins with importing some necessary libraries and dataset to Google Colab. Next, our team performs data exploration and analysis to examine the characteristics of each attribute, trends or relationships between variables, and between the target variable and other variables in the dataset. We also check for null data, duplicated columns and delete some unnecessary columns in the pre-processing stage. The other columns will be used for each model to produce the final prediction results for the entire dataset. Details for each step are explained in the following sections.

3.3. The data source

The research team has selected the Bank Customer Churn dataset, which has a decent amount of diverse data attributes related to Customer Churn activities. In other words, the dataset is suitable for building a meaningful model. The dataset includes information about customers with 12 columns and 10,000 rows from ABC Multinational bank.

The dataset includes personal information of customers identified by `customer_id`. The target variable is "churn" which is classified as "churn = 1" when the customer

leaves and "churn = 0" when the customer does not leave. To understand more about the dataset, a data description table based on ABC's explanation is provided in Table 2 below:

Customer Identity	
Attribute	Description
customer_id	Account Number
credit_score	Credit Score of customer
country	Country of Residence
gender	Sex
age	Age of customer
tenure	From how many years he/she is having bank account
balance	Account Balance
products_number	Number of Product from bank
credit_card	Does this customer have a credit card ?
active_member	Is he/she an active Member of the bank ?
estimated_salary	Salary of Account holder
churn	Churn Status. 1 if the client has left the bank during some period or 0 if he/she has not.

Table 2: Data description

(Resource: Website kaggle.com about ABC Multistate Bank)

3.4. Exploratory data analysis

As we can see in the heat map that the variables have strong correlation (threshold:0.8). Features show no strong correlations as the chart showed. So there are still no correlations within the churned customer dataset; as it was with the total customer dataset.

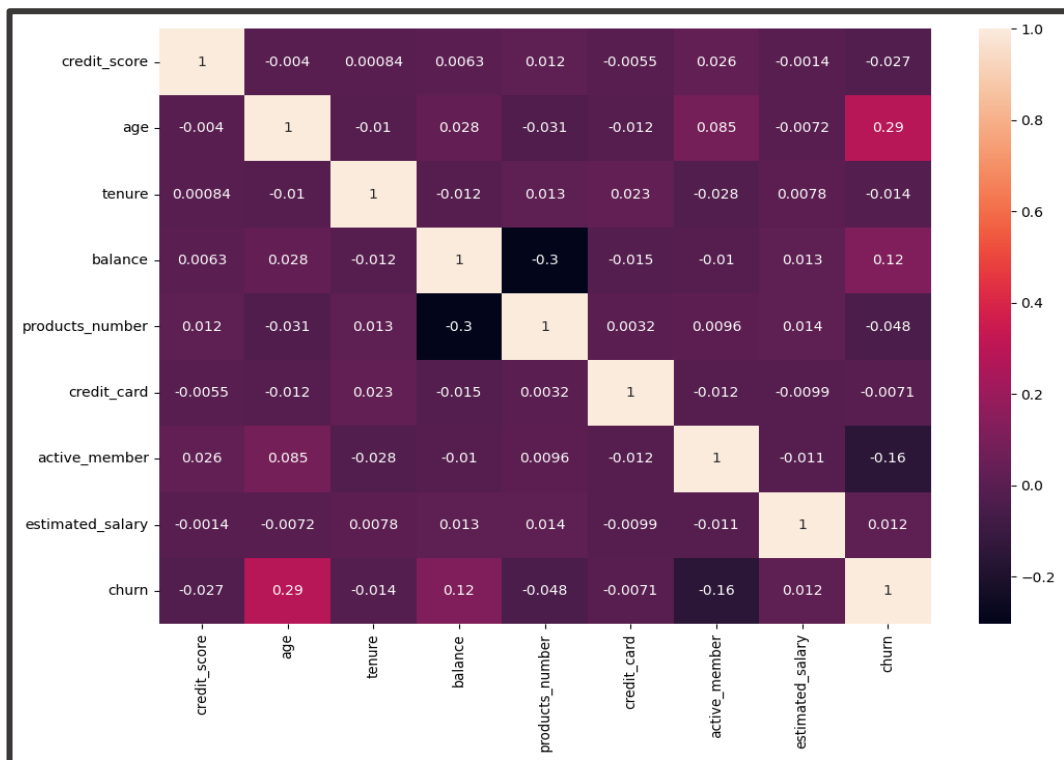


Figure 8: Heatmap correlation between variables

(Resource: Do-it-yourself team)

We will explore the numerical features first: credit_score, age, tenure, products_number, balance, estimated_salary. Notice that, even some types of features are numerical types but the values indicated are binary (0 or 1). So, I consider these features as categorical data.

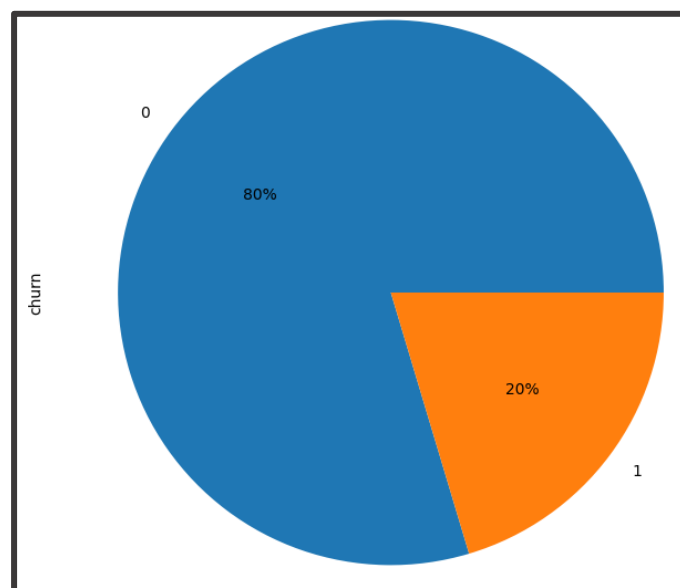


Figure 9: Pie chart of churn and not churn customer

(Resource: Do-it-yourself team)

The bank managed to retain 80% of their customers with the remaining 20% churned out. So that the dataset is imbalanced.

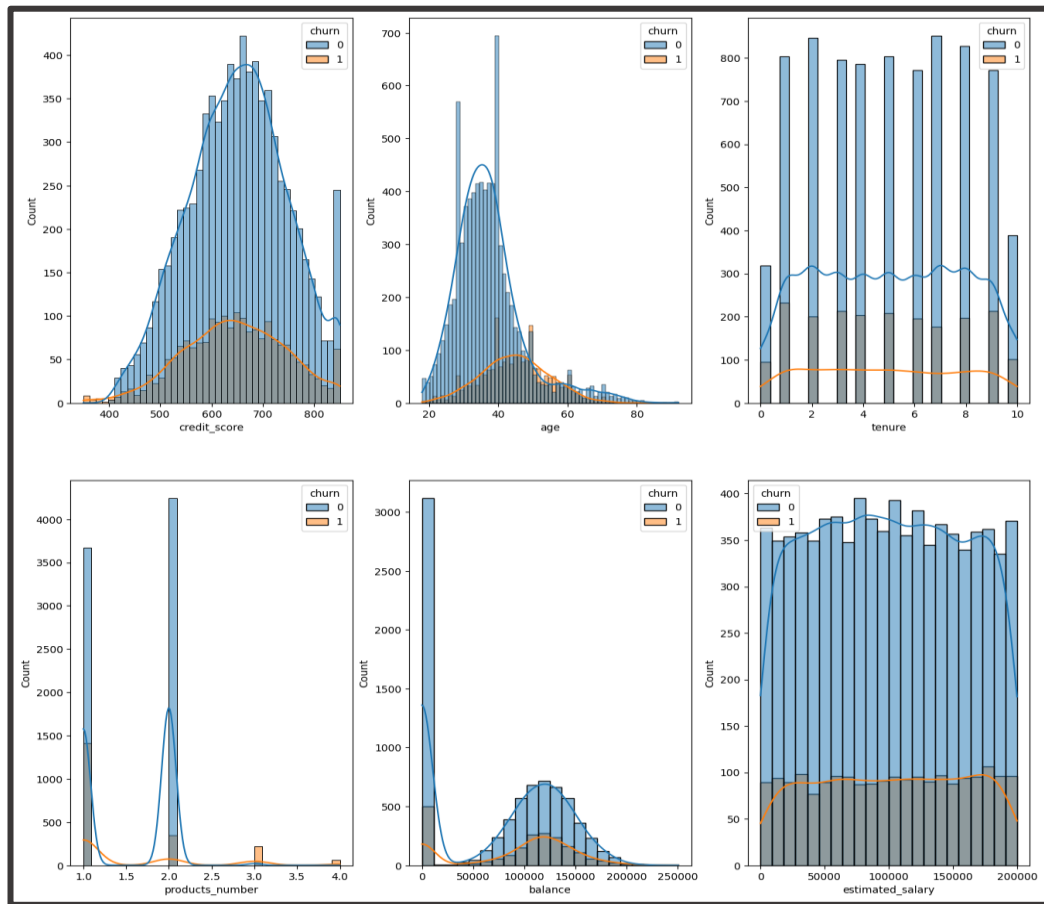


Figure 10: Customer and Churn customer analysis

(Resource: Do-it-yourself team)

3.4.1. Customer analysis

The contributions above show that the average credit scores of the customers is **650**, with most of the customers having credit scores **between 600 and 700**. Majority of the bank's customers are below **the age of 50**; and have the average **age of 39**.

3.4.2. Churned customer analysis

The average credit score for the churned customers (**645**) is slightly lower than the overall average credit score, but the majority of the credit scores of churned customers still falls **between the 600 and 700** range. The average age of the churned customer (**45**) is higher than that of the retained customer (**37**) with the majority of the churned customers **between the ages of 35 and 60**. Majority of the bank's churned customers

left after spending **one year** at the bank. Customers that purchased more products through the bank tended to be churned as **100% of the customers** that purchased 4 products left and **79%** of those that purchased 3 products also were churned out.

Due to the unbalanced dataset. It is hard to come to a conclusion. But the charts indicate that **credit_score, balance, tenure and estimated_salary show no differences with the number of customers who churn or not as the distribution of both kinds of these features overlapping**. With products_number and age, there is a point, most of the customers who churn have bought more than 2 products from the banks while most of the customers are around 20 to 50. However, the dataset is imbalanced. So I will perform inferential analytics on all these features to have a better view.

```
p-value = 0.01, alpha = 0.05
p < alpha => reject H0 => there are differences between credit_score and churn

p-value = 0.00, alpha = 0.05
p < alpha => reject H0 => there are differences between age and churn

p-value = 0.17, alpha = 0.05
p > alpha => failed to reject H0 => there are NO differences between tenure and churn

p-value = 0.00, alpha = 0.05
p < alpha => reject H0 => there are differences between products_number and churn

p-value = 0.00, alpha = 0.05
p < alpha => reject H0 => there are differences between balance and churn

p-value = 0.23, alpha = 0.05
p > alpha => failed to reject H0 => there are NO differences between estimated_salary and churn
```

Figure 11: Welch's t-test indicates

(Resource: Do-it-yourself team)

As the test indicates, tenure and estimated_salary failed to reject H0. So they do not affect the churning while others show effect. From here, We will remove these features from the dataset.

The bank has a large customer base from France; with more than **50% of the customers banking from France**. A huge percent of the bank's customers are inactive; **48%** of the total customer count are reported to be inactive members. **Only 29%** of the ABC multinational bank's customer base do not make use of credit cards. Germany recorded the highest number of churned customers with **48% of the German customers** churned. Majority of **the churned customers (56%)** were female despite the total

population of the bank being predominantly male. As expected most of the customers who left the bank were inactive members which could serve as an indicator as to why they left. But **36%** of the churned customers were still active customers.

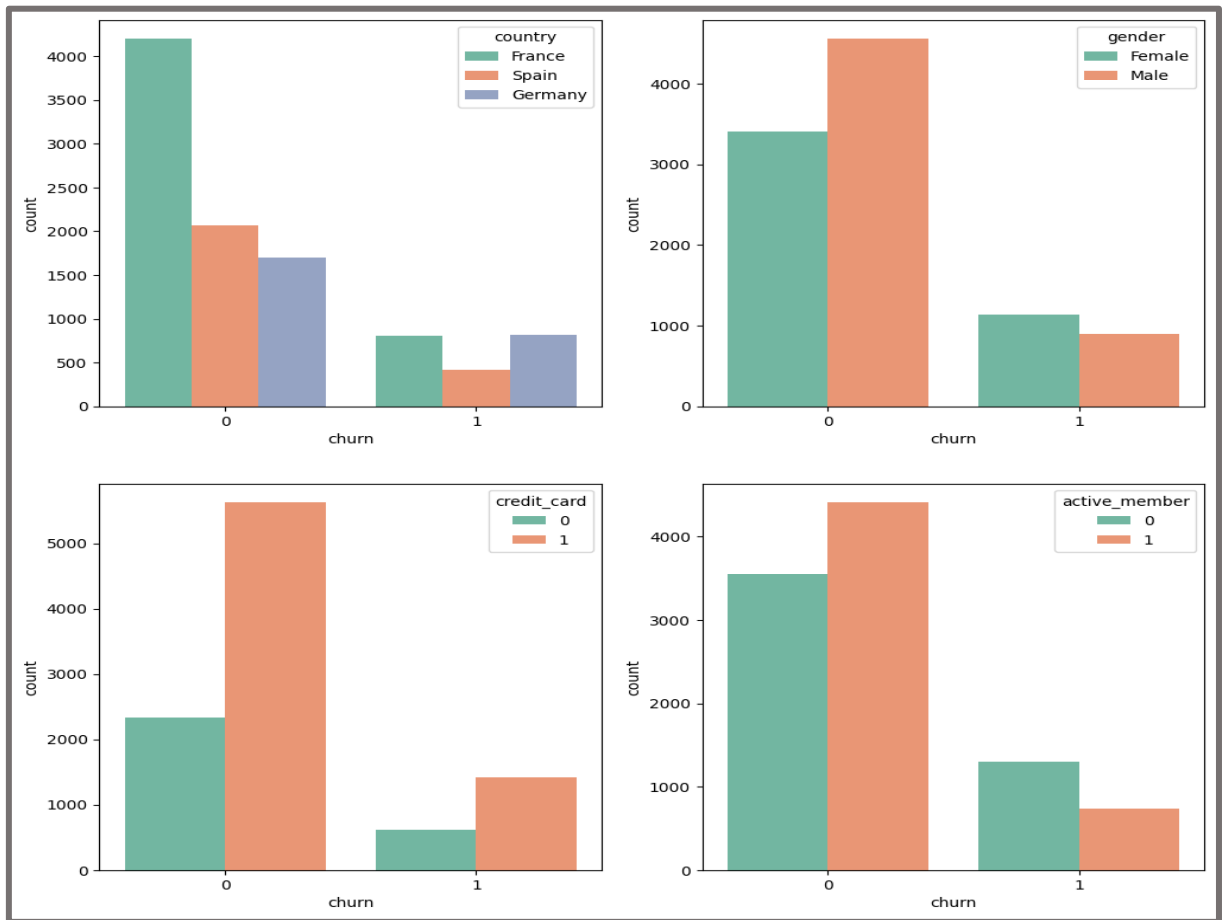


Figure 12: Total number of customers who have churned by country, gender, credit card and active member.

(Resource: Do-it-yourself team)

All features affect the churn decision of customers except for credit_card. The trend of people who have a credit card and decided to churn or people who don't have a credit card but decided to churn are no different and vice versa. We will perform the inferential analytics on these features so I can come to the conclusion.

```

p-value = 0.00, alpha = 0.05
p < alpha => reject H0 => country and churn are DEPENDENT

p-value = 0.00, alpha = 0.05
p < alpha => reject H0 => gender and churn are DEPENDENT

p-value = 0.49, alpha = 0.05
p > alpha => failed to reject H0 => credit_card and churn are INDEPENDENT

p-value = 0.00, alpha = 0.05
p < alpha => reject H0 => active_member and churn are DEPENDENT

```

Figure 13: Chi Square Test

(Resource: Do-it-yourself team)

As I doubted, credit_card and churn are independent. So there are no differences about the churn condition among people who have a credit card or not. While other features show a correlation with churn condition. I delete the credit_card column from the data to remove the unnecessary column for the model.

3.5. Data pre-processing

Our dataset is quite clean so we just use a few steps to pre-processing this data. We check for shape and info data. Also check for nulls and duplicate values. After watching all the columns, we decided to drop customer_id because it is unnecessary.

```

RangeIndex: 10000 entries, 0 to 9999
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           10000 non-null   int64
1   credit_score           10000 non-null   int64
2   country                10000 non-null   object
3   gender                 10000 non-null   object
4   age                   10000 non-null   int64
5   tenure                 10000 non-null   int64
6   balance                10000 non-null   float64
7   products_number       10000 non-null   int64
8   credit_card           10000 non-null   int64
9   active_member          10000 non-null   int64
10  estimated_salary       10000 non-null   float64
11  churn                  10000 non-null   int64
dtypes: float64(2), int64(8), object(2)
memory usage: 937.6+ KB

```

Figure 14: Data columns

(Resource: Do-it-yourself team)

CHAPTER 4 : EMPIRICAL AND PROPOSED RESULTS.

In this chapter, the research team evaluates and discusses the accuracy and results of the model using metrics applied to the classification problem.

4.1 Experimental results

Results of applying the cross validation in all models are shown in Table 3 and Figure 15.

LR	KNN	RF	CART	XGB	SVM
0.679	0.7495	0.8335	0.7245	0.8115	0.7835
0.679	0.7495	0.8375	0.7245	0.8115	0.7835
0.684	0.7415	0.8225	0.7265	0.8185	0.783
0.655	0.5875	0.8505	0.756	0.8255	0.459
0.712	0.7605	0.85	0.7715	0.8255	0.785
0.7185	0.7685	0.851	0.753	0.8325	0.7995
0.692	0.7655	0.8415	0.734	0.815	0.7785
0.697	0.758	0.8415	0.7365	0.8135	0.786
0.684	0.765	0.8445	0.7265	0.822	0.7805
0.7075	0.7595	0.835	0.6895	0.829	0.7815

Table 3: Accuracy of models during 10 folds

(Resource: Do-it-yourself team)

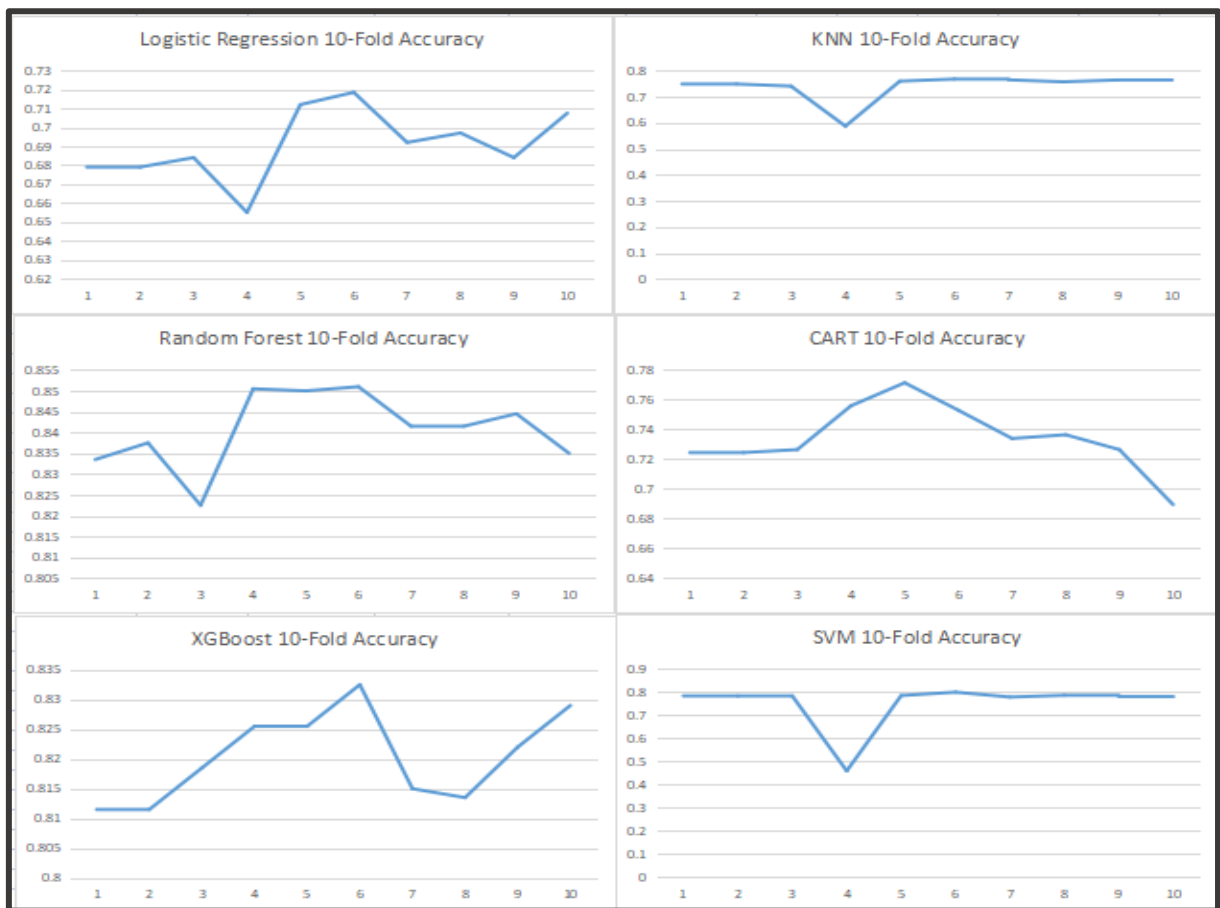


Figure 15: 10-Fold accuracy of the selected models

(Resource: Do-it-yourself team)

Minimum and maximum accuracies for all of the selected models are summarized in Figure 15. Results of the study show that ensemble based learning techniques (RF and XGBoost) achieved the highest performance with approximately 86%. Both CART and SVM can be recommended as well with 77% and 80% accuracy. KNN 76% and finally LR with accuracy 71% as shown in Table 4.

Model	Min	Max
Logistic Regression	0.655	0.7185
KNN	0.5875	0.7685
Random Forest	0.8225	0.851
CART	0.6895	0.7715
XGBoost	0.8115	0.8325
SVM	0.459	0.7995

Table 4: Accuracy of the selected models

(Resource: Do-it-yourself team)

4.1.1. Stacking model

Now, we perform the stacking model to see if it can improve the performance.

To examine the performance of the classification algorithms, we utilized a variety of performance measures. Accuracy, precision, recall, sensitivity, and the ROC curve are the measures in question. The accuracy metric indicates the proportion of properly classified instances; however, it is insufficient for evaluating the classifier's performance. Table 5 show the overall performance of the base model, while Table 6 presents the performance of different meta-models.

Level 1 - Base model classifier	Performance Measurement			
	Accuracy	Precision	Recall	ROC
LogisticRegression	0.7	0.89	0.7	0.68
KNeighborsClassifier	0.74	0.91	0.75	0.74
RandomForestClassifier	0.845	0.88	0.92	0.73
DecisionTreeClassifier	0.77	0.9	0.79	0.73
XGBClassifier	0.829	0.9	0.88	0.76
SVC	0.787	0.92	0.8	0.76

Table 5: The overall performance of base-model classifiers

(Resource: Do-it-yourself team)

Based on Table 5, we could notice that the best base model is Random Forest with an accuracy of 84.5%, precision of 88%, recall of 92% and ROC of 73%. The XGB Classifier, on the other hand, has an amazing ROC performance of 76%, but its accuracy of 82.9% is somewhat lower than that of the Random Forest.

Another finding in Table 5 is that there are few single classifiers with low accuracies, such as the Logistic Regression (with a 70% accuracy), perform poorly compared to other classifiers. The same low performance is also revealed by KNeighborsClassifier classifiers. Therefore, these ensembles included several kinds of models, all of which provide a desirable degree of variety when it comes to predictions. Consequently, we may suppose that our base-model pool is made up of both excellent and bad classifiers and that the rule of meta-model at the next level is to merge them in order to create a superior model.

Level 2 - Meta-model classifier	Performance Measurement			
	Accuracy	Precision	Recall	ROC
Logistic Regression	0.88	0.9	0.95	0.7981
KNeighborsClassifier	0.85	0.87	0.96	0.7068
RandomForestClassifier	0.845	0.88	0.94	0.703
DecisionTreeClassifier	0.8625	0.86	0.99	0.69
XGBClassifier	0.85	0.89	0.93	0.7252
SVC	0.86	0.87	0.97	0.7252

Table 6: The overall performance of level 2 - meta-model classifiers

(Resource: Do-it-yourself team)

Based on Table 5, we have developed, tested and compared with the six base-model classifiers: Logistic Regression, KNeighborsClassifier, Random Forest, DecisionTreeClassifier, XGB Classifier SupportVectorMachine, our proposed stacking ensemble classifier has achieved excellent classification results. All meta-models of stacking ensemble classifiers gained significantly better performance than individual classifiers and voting model.

The Stacking Ensemble (RandomForest) is the weakest meta-model, with an accuracy of 84.5%. According to Table 5, the Stacking Ensemble (LogisticRegression) meta-model classifier surpassed all other models with a classification accuracy of 88.0%. In addition, it had the highest ROC of 79.8%. Stacking Ensemble (Decision Tree) performance might also be considered since it attained almost the same accuracy (86.25%) as the top meta-model.

The proposed stacking ensemble method to classify customer churn has a high performance since the base classifiers are stacked, combining their predictive power. Different classifiers in this model compensate for the shortcomings of other classifiers, resulting in an overall improvement in performance. The suggested stacking ensemble is a one-of-a-kind combination of heterogeneous base classifiers and meta classifiers that perform best at classification.

4.1.2. Voting model

Besides that, we detail our hybrid classification via a voting-based approach. We consider soft voting for our model. We use the classifiers such as the Logistic

Regression, KNeighborsClassifier, RandomForestClassifier, DecisionTreeClassifier, XGBClassifier and SVM as an ensemble combination within our voting-based approach.

After running voting models to test the accuracy of models, our performance is on the range with min accuracy 83.8% to 84.9%, AUC of 0.86, with precision and recall of 0.92 and 0.88, respectively. This result is quite better than single models.

4.2 Discussion

With high accuracy and good results obtained from Voting and Stacking models, the effectiveness of these models becomes limited when dealing with highly imbalanced datasets like the one used to build the model. Because the model's predicted results will be biased towards the majority class due to the large number of customers who do not churn, it heavily influences the model's predictive ability. As a result, the model is highly likely to predict that all customers will not churn. Therefore, to determine the model's predictions and make them applicable in practice, relying solely on accuracy is not enough.

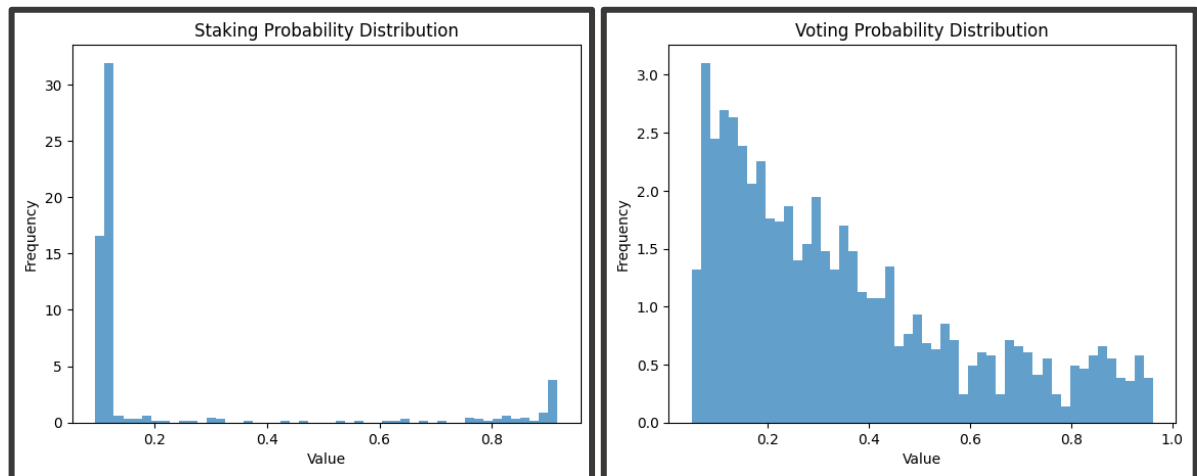


Figure 16: The frequency distribution graph of bank churning customers

(Resource: Do-it-yourself team)

The distribution graph in Figure 12 shows that the higher the probability of users churning within the range of $[0, 0.1]$ for the Stacking model and $[0, 0.5]$ for the Voting model, the higher the likelihood of churning. Specifically, for the results obtained from Figure 16, the threshold must be a number greater than or equal to 0.2 and as close to

0.2 or 0.5 and as close to 0.5 as possible for each criterion. To find this threshold, the research team evaluated the effectiveness of the model based on two criteria as follows:

Criteria 1: ROC-AUC curve

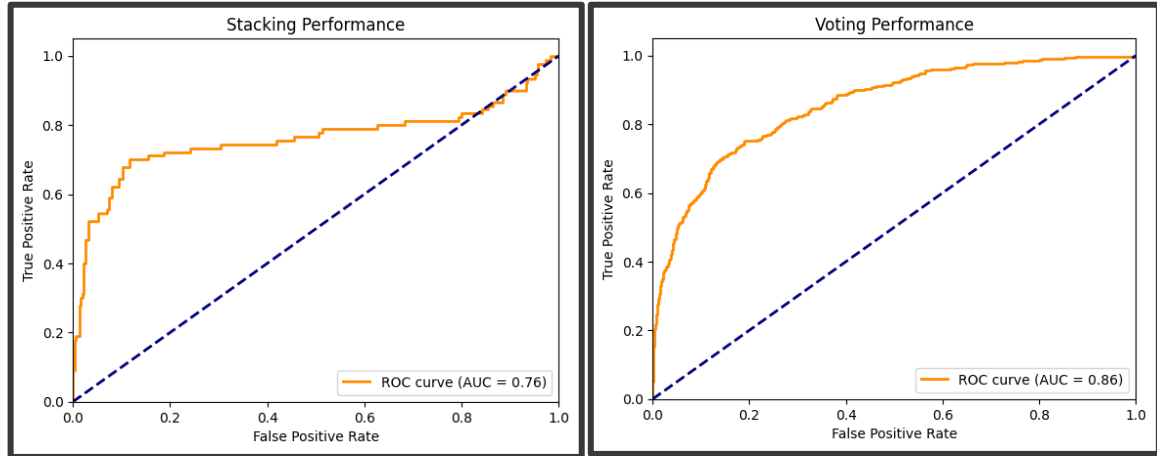


Figure 17: ROC-AUC curve

(Resource: Do-it-yourself team)

As emphasized, the closer the curve is to the point (0, 1), the more effective the model is. Therefore, the Voting model provides quite good predictions, as confirmed by an AUC of 0.86. Meanwhile, the Stacking model has a smaller area under the ROC curve, but the difference is not significant (AUC = 0.76).

Criteria 2: Precision-Recall curve

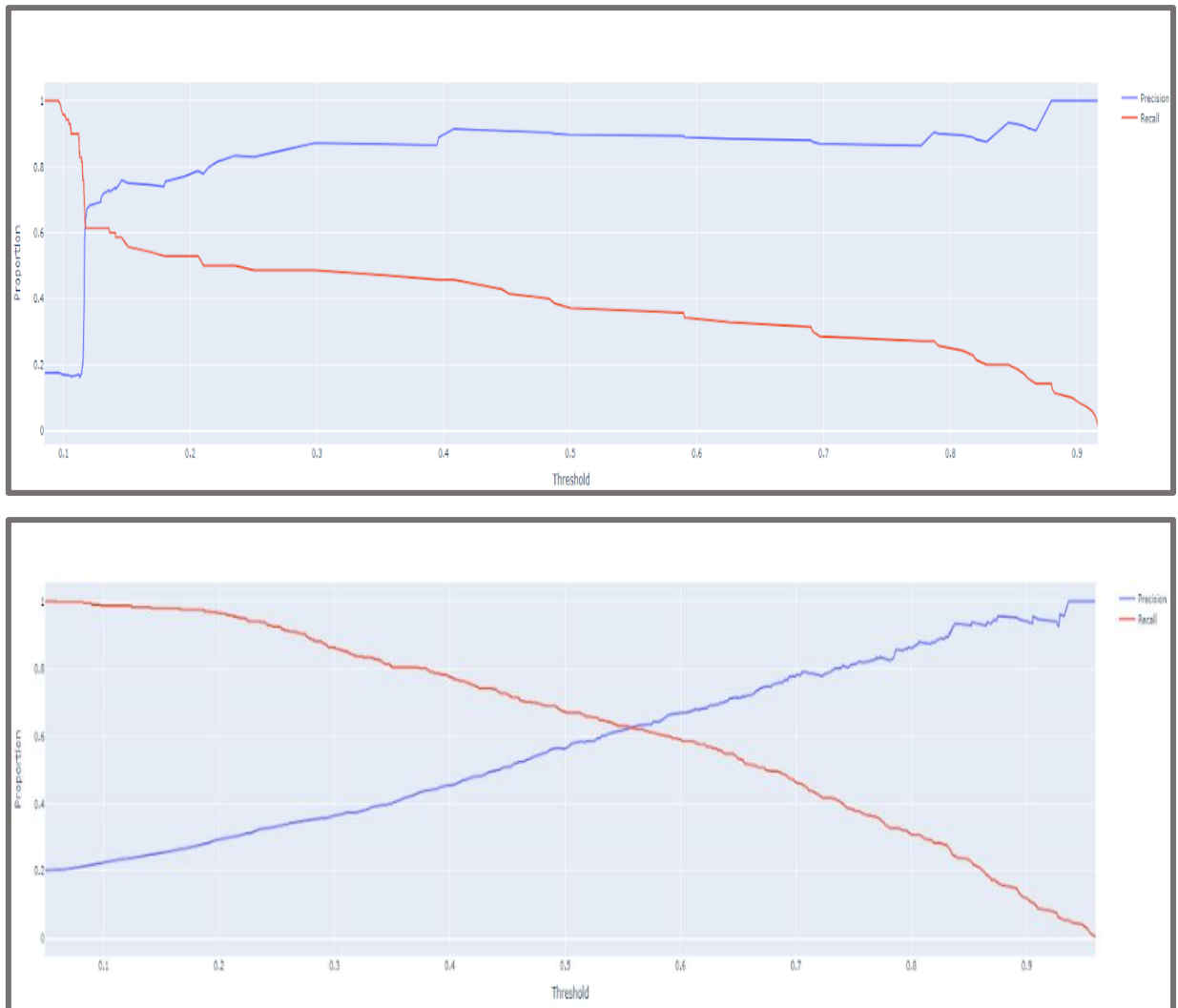


Figure 18: Precision and Recall curve

(Resource: Do-it-yourself team)

The Precision and Recall curves intersect at a threshold of 0.1 and 0.55 with Stacking and Voting in Figure 17, indicating that the accuracy in predicting churning customers with a probability greater than or equal to this threshold is very high. This is consistent with the frequency distribution plot in Figure 18.

4.3 Proposals

Based on the analyst from the ABC Multinational Bank dataset we have some recommendation grouped by category:

4.3.1. Country

The French branches make up a huge portion of the customer base.

The bank should set up a campaign to expand the customer base in Germany, Spain, and could later look to expand to other surrounding countries.

German branches recorded the highest churn rate with 48% hence further analysis would be required to determine why.

4.3.2. Credit Scores

The credit scores of most customers of the bank are generally considered to be above the good threshold.

Notwithstanding, good credit scores were not enough to deter customers from churning out.

4.3.3. Gender

The gender split between the male and female customers was 55% and 45% respectively.

Most of the churned customers turned out to be women so further analysis is required to find out the cause.

4.3.4. Activity Levels

A whopping 48% of the bank's customers were Inactive which is far from ideal; hence, analysis would be required to determine the cause of the high percentage of inactive members.

4.3.5. Age

Younger customers made up most of the retained customer base so efforts should be made to acquire young customers as they were less likely to leave.

Analysis should also be carried out to determine why older customers were more likely to leave.

4.3.6. Tenure and Purchased Products

Majority of the churned customers left after one year with the bank; although it is important to note that customers still left after spending a considerable amount of time at the bank.

Analysis is required to determine why customers that bought more products with the bank ended up leaving the bank, as none of the customers that bought 4 products with the bank were retained.

4.3.7. Churn Rate

The bank recorded a churn rate of 20% which is below the average churn rate for financial institutions which is 25%. Hence, The bank should keep implementing more methods and strategies to reduce the churn rate even further.

CHAPTER 5 : CONCLUSION AND FUTURE WORKS

5.1. Conclusion

Various machine-learning techniques have been used for customer churn in CRM. This study proposes a customer-churn prediction system based on the stacking and voting ensemble of machine learning. Feature construction is used to expand the feature space and discover latent information from implicit features. The proposed feature construction through feature grouping improves the prediction accuracy compared to the original customer-churn dataset. The result of proposed system showed the best accuracy of 84.5% (Random Forest) and 88% (Logistic Regression), precision of 92% (SVC) and 90% (Logistic Regression) , recall of 92% (Random Forest) and 99% (Decision Tree), ROC of 76% (both XGBoost and SVC) and about 80% (Logistic Regression) for the old and new datasets respectively compared to other prediction systems. This proposed system can determine important factors affecting customer purchasing behavior in the finance - banking industry.

The research topic is “*Machine Learning Techniques for Customer Churn: A comparative Study*”. Building a model to predict customer churn rate at businesses and companies,... to improve the accuracy of predicting the future of customer attitudes, behaviors, and habits leading to churn. This can be a part of contributing to the improvement and enhancement of the system of the business because during this period, , most people are aiming for good, safe, and beneficial things for themselves, so it is necessary to understand customer psychology to offer better types of services and products to retain loyal customers and attract to new ones.

The proposed model is built on machine learning algorithms and applied on datasets with the same size and properties as the actual dataset. The research project has achieved the following results:

Firstly, the research team has a better understanding of methods to predict customer churn rates, this is one of the issues receiving great attention of the Finance – Banking industry in particular and other related industries and fields in general. In the process of learning, the team discovered that there are many algorithms to build analytical models, predictions, as well as having many models combined from other algorithms yield better

results. So this paper has focused on building on the characteristic, properties of the dataset to select the most suitable algorithms.

Secondly, the research team proposed a combined model between algorithms to build the model, then make reviews, comments. The evaluation results of the model show that the prediction accuracy is higher than that of the single algorithmic model, The model can be used integrally in software applications to predict customer churn rates in businesses.

5.2. Restriction

During the implementation of the project, due to the scarcity of time, human resources as well as knowledge are not yet specialized in the field, so, the research team still has many limitations, specifically including:

Firstly, the analysis model time is large, the speed of giving results is not optimal.

Second, this study only builds an improved model to accurately predict churn rates. However, there will be many cases of customers with psychology, habits, behaviors lead to the model not being able to make completely accurate predictions, but only making predictions at the optimal level of the model. Therefore, businesses need to coordinate many tools, models and methods to increase the rate of accurate predictions higher to be able to capture the psychology of customers, avoid leading to customer churn. Moreover, this can also be a positive development direction for other researches in the future.

5.3. Future works

This is a matter of concern in the current 4.0 technology era, so the research topic can be developed and applied in many different aspects, the research team has proposals for future research expansion, including:

First, optimizing models with methods capable of increasing model performance, reducing analysis time and classification.

Second, improve the model to help businesses more accurately predict customer churn rate (that is, correctly predict customers with churn signs) to increase profits,

reduce loss costs for attracting new customers equal to the number of customers churned.

Third, form the inclusion of a model that distinguishes between loyal and non-loyal customers. Various researchers have documented the current limitations of a churn management solution. Therefore, there is a need to develop a framework for calculating “loyalty index”.

Fourth, provide the basis for future work in several directions:

- Individual models have been researched extensively over the years and the most relevant final predictions usually be unstable (can well or no-well). While we used stacking model combine the predictions of multiple models to produce a final prediction that is more accurate than any individual model. This will aim to result in better prediction accuracy for customer churn is necessary. It will also be of interest to help businesses identify customers who are likely to churn and take appropriate action.
- Another interesting alley for research is the use of the proposed model in additional models and applications. Because it can flexible and adaptable, which makes them suitable for wide range of data science problems. It can be used with different types of models and algorithms, and can be customized to suit the specific needs of a business or application. For example, the method can be used for campaign management, by creating the best prediction model to be approached with a specific marketing campaign.
- Finally, Our method can also be used in other domains because stacking model are transparent, which means it is possible to understand how the final prediction is generated by looking at the individual models that were used. This can help businesses to gain insights into the factors that are driving customer churn and develop strategies to address them.

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APPENDIX

1. Google Colab:

https://colab.research.google.com/drive/1j98tFmdX3rgHSWfedouI40w2gDqzrj9P#scrollTo=MakduiH6Erl_

2. Stacking model:

```
lr_model = LogisticRegression()
# lr_model.fit(X_train, y_train)
# y_pred_lr = lr_model.predict_proba(X_test)

knn_model = KNeighborsClassifier(n_neighbors=11)
knn_model.fit(X_train, y_train)
y_pred_knn = knn_model.predict_proba(X_test)

rf_model = RandomForestClassifier()
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict_proba(X_test)

dt_model = DecisionTreeClassifier(max_depth=3)
dt_model.fit(X_train, y_train)
y_pred_dt = dt_model.predict_proba(X_test)

xgb_model = XGBClassifier()
xgb_model.fit(X_train, y_train)
y_pred_xgb = xgb_model.predict_proba(X_test)

svm_model = SVC(probability=True)
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict_proba(X_test)

df_meta = pd.DataFrame()
# Linear Regression
# y_pred_lr = pd.DataFrame(y_pred_lr)
# df_meta['data_lr_0'] = y_pred_lr[0]
# df_meta['data_lr_1'] = y_pred_lr[1]
```

```

#-----
# KNN
y_pred_knn = pd.DataFrame(y_pred_knn)
df_meta['data_knn_0'] = y_pred_knn[0]
df_meta['data_knn_1'] = y_pred_knn[1]
#-----

# Random Forest
y_pred_rf = pd.DataFrame(y_pred_rf)
df_meta['data_rf_0'] = y_pred_rf[0]
df_meta['data_rf_1'] = y_pred_rf[1]
#-----

# Decision Trees
y_pred_dt = pd.DataFrame(y_pred_dt)
df_meta['data_dt_0'] = y_pred_dt[0]
df_meta['data_dt_1'] = y_pred_dt[1]
#-----

# XGB
y_pred_xgb = pd.DataFrame(y_pred_xgb)
df_meta['data_xgb_0'] = y_pred_xgb[0]
df_meta['data_xgb_1'] = y_pred_xgb[1]
#-----

# SVM
y_pred_svm = pd.DataFrame(y_pred_svm)
df_meta['data_svm_0'] = y_pred_svm[0]
df_meta['data_svm_1'] = y_pred_svm[1]
#-----

df_meta['data_test'] = y_test.reset_index().drop(columns=['index'])
df_meta.head()

X_ = df_meta.drop(columns=['data_test'])
y_ = df_meta['data_test']

X_train_, X_test_, y_train_, y_test_ = train_test_split(X_, y_, test_size=0.2)

```

```

meta_model = LogisticRegression()
meta_model.fit(X_train_, y_train_)
y_pred_meta_ = meta_model.predict(X_test_)
print('accuracy:', accuracy_score(y_test_, y_pred_meta_))

cm = classification_report(y_test_, y_pred_meta_)
print(cm, '\n')

```

3. Voting model:

```

lr_model_voting = LogisticRegression()

knn_model_voting = KNeighborsClassifier(n_neighbors=11)
rf_model_voting = RandomForestClassifier()
dt_model_voting = DecisionTreeClassifier(max_depth=3)
xgb_model_voting = XGBClassifier()
svm_model_voting = SVC(probability=True)

voting_clf = VotingClassifier(estimators=[('lr', lr_model_voting), ('knn',
knn_model_voting), ('rf', rf_model_voting), ('dt', dt_model_voting), ('xgb',
xgb_model_voting), ('svm', svm_model_voting)], voting='soft')

voting_clf.fit(X_train, y_train)

y_pred_voting = voting_clf.predict(X_test)
print('accuracy:', accuracy_score(y_test, y_pred_voting))

cm_voting = classification_report(y_test, y_pred_voting)
print(cm_voting, '\n')

```